



Engineering Capstone Project Part A

Creating a real-time digital twin of a truss bridge

Project Proposal

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Summary:

This project investigates the viability and layout of a real-time digital twin of a laboratory ≈ 2 m truss bridge inside a MATLAB setup. The digital twin integrates real-time sensor measurements, which are primarily strain, acceleration, and displacement into an approximate physics model that determines member axial actions (tension/compression) and global deflection in real time. The objectives of Capstone A are to (i) define a sensor placement and configuration that optimises information content on truss behaviour, (ii) specify a data pipeline and estimation procedures suitable for low-latency real-time usage, and (iii) develop methods to calibrate and validate the twin against controlled laboratory measurements. The suggested work is founded on documented research in structural health monitoring (SHM), optimal sensor placement, and model updating [4–6,8–10].



Figure 1. 2m truss bridge at the RMIT lab

Source: Professor Ricky Chan

Statement of Problem:

Bridges are safety-critical infrastructure. Traditional regimes of inspection are periodic and reactive, permitting undetected degradation between visits. SHM modifies this to continuous evidence, yet many systems remain sensor-based (alarms/dashboards) without an integrated physics-based framework. A digital twin, which is a synchronised dynamic digital replica of the physical structure enables real-time estimation of unmeasured states (member forces, stiffness variation) and forward prediction of serviceability issues [1–3]. With truss bridges, where axial load and connection conditions dominate behaviour, the engineering questions are practical and specific: what sensors to employ, where to locate them, and how to link data to a model in real time such that the twin becomes reliable after calibration [1–3,4–6,8–11]. This project addresses these questions using a controlled laboratory truss for repeatable loading and verification.

Background and Literature Review

1. Digital Twins and SHM for Civil Structures

A digital twin is a cyber-physical representation that unites a physics model with live data to facilitate diagnostics and prognostics across an asset's life [1–3]. In civil infrastructure, twins bring to SHM a closed-loop model-updating paradigm, which is continually balancing predictions and measurements and quantifying uncertainty [3,4,10]. In practice, the twin includes: (i) a physical core (e.g., FE truss model), (ii) a sensing layer (strains, accelerations, displacements), (iii) estimation algorithms (e.g., Kalman filters, recursive least squares), and (iv) decision logic and visualisation [4,10–13].

2. Truss Behaviour and Measurable Quantities

In a pin-jointed truss, members primarily resist axial tension or compression; serviceability is governed by deflection and vibration response, while safety checks include yield, buckling (for compression members), and connection/slip [7,17]. Local force-related data are most directly obtained from member strain; accelerations provide global dynamic properties (frequencies, mode shapes); displacements measure global stiffness and boundary conditions [4–7].

3. Sensor Technologies and Practicalities

Foil strain gauges provide low-cost, high-accuracy strain measurement with careful surface preparation, temperature compensation, and adhesion [7]. MEMS accelerometers capture global dynamics (wired or wireless), for which time synchronisation and sampling strategy matter [6]. LVDTs/laser displacement sensors provide accurate deflection at mid-span or supports [7]. A practical laboratory arrangement is mixed: concentrated strain gauges on selected members, 1–2 accelerometers at mid-span/quarter points, and one displacement channel at mid-span for overall stiffness calibration [4,6,7].

4. Optimal Sensor Placement (OSP)

OSP favours sites that maximise state/parameter information. Effective Independence and Fisher-information criteria are common (optimising mode-shape observability or parameter sensitivity) [8,9]. For trusses, engineering intuition complements theory: strain gauges at bottom chord mid-span (high tension from gravity), diagonals with high panel forces, and critical joints; accelerometers at large global mode amplitudes (mid-span/quarter points) [4,8,9,12]. Redundancy increases resilience to dropouts and drift.

5. Model Updating and Real-Time Estimation

Model updating adjusts parameters (member stiffness, boundary conditions) to bring predictions and measurements into agreement using static or dynamic data [10]. Kalman-filter families are used for real-time state estimation with noise and missing data, combining prediction and measurement for minimum-variance estimates [11]. For a small truss with quasi-static loading, a discrete-time Kalman filter for displacements/velocities plus recursive least squares for stiffness scalars is effective and computationally efficient [10–12]. Operational modal analysis supports validation without controlled excitation [12].

6. Failure-Point Inference

A calibrated twin supports “what-if” scenarios to identify critical members under moving loads via influence lines and to assess yield/buckling margins using code-consistent checks [7,17]. Member-level force histories from the twin can be compared against allowable demands to flag potential failure-point candidates for closer inspection and test confirmation.

7. Prior Art and Gaps Filled

The literature consistently shows (i) advantages of multi-sensor SHM, (ii) efficacy of update strategies, and (iii) potential of digital twins; fewer studies describe low-latency, lab-scale deployments with explicit placement rationale and closed-loop calibration reproducible within a semester [1–3,6,8–11]. This work addresses that gap by specifying a conservative sensor suite, a MATLAB pipeline, and testable calibration/validation protocols on a ≈ 2 m truss.

List of Design/Research Questions:

1. Which sensor types and placements best capture, in real time, the structural behaviour of the laboratory truss bridge?
2. How should a real-time digital twin of the truss bridge be designed (architecture, physics model, estimation, and interfaces)?
3. How can the digital twin be implemented in MATLAB to ingest live sensor data for structural health monitoring?
4. To what extent can a calibrated twin identify likely failure points in the truss under laboratory loading?

Methodology:

A. Time Planning (26 Weeks; Capstone A = Weeks 1–13, Capstone B = Weeks 14–26)

Weeks 1–2 (A): Requirements & literature: confirm lab access and safe load envelopes; specify performance targets (e.g., 5–10 Hz update; $\leq 10\%$ RMS strain/deflection error at service loads); review SHM/OSP/model-updating methods [4–6,8–13].

Weeks 3–4 (A): Concept & sensor plan: choose sensor suite (strain gauges, 1–2 accelerometers, 1 mid-span displacement channel) and draft placement map from simple FE influence lines and expected mode shapes [7–9,12].

Weeks 5–7 (A): Pipeline scaffolding: implement MATLAB acquisition broker (timestamping, QA checks), 2-D truss solver (analytical/FE), and estimation kernel (discrete Kalman + recursive least squares); define a latency budget across acquisition/filtering/solve/display.

Week 8 (A): Calibration protocols: plan static load steps and light dynamic taps (for OMA); define metrics (strain/deflection residuals, frequency error, localisation score) [10–12].

Weeks 9–11 (A): Dry-runs & tuning: synthetic-data tests then limited lab trials at moderate loads; tune noise covariances, outlier removal, dropout handling [6,11].

Week 12 (A): Documentation: results, limitations, and Part-B plan; lab SOPs (sensor bonding, DAQ checks, safe loading).

Week 13 (A): Finalise and submit Capstone A report.

Weeks 14–15 (B): Implementation hardening: refine model assembly, improve UI/visualisation, add robustness to sensor dropouts (hold-last-good, re-initialisation).

Weeks 16–18 (B): Integrated testing with live data: extended runs under varied load paths; collect datasets for validation.

Weeks 19–21 (B): Validation & sensitivity: hold-out experiments; emulate damage via virtual stiffness reductions; evaluate failure-point inference (precision/recall vs labelled members).

Weeks 22–24 (B): Model refinement: address discrepancies; revisit sensor placement if needed; finalise acceptance criteria and decision thresholds.

Week 25 (B): Final reporting draft and presentation build: figures, results, discussion aligned to CLOs.

Week 26 (B): Final report submission and presentation delivery.

B. Resource Planning

Figure 2 summarises the staged plan across Capstone A and Capstone B.

Capstone A + B: 26-week Gantt

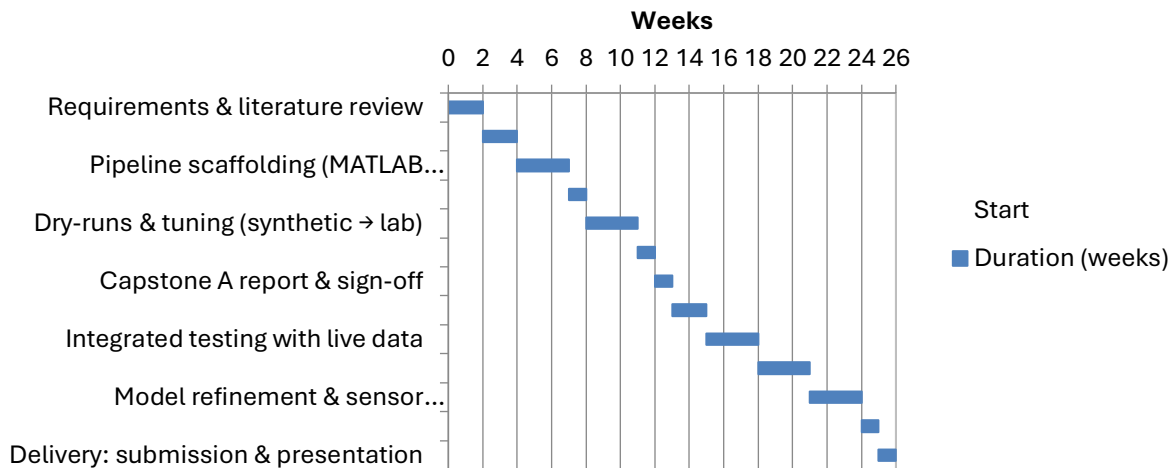


Figure 2. 26-week schedule for Capstone A

B. Resource Planning

People: Team of five (Leads: Sensing, Modelling, Software, QA/Validation, Documentation). Weekly supervisor consults.

Hardware: Strain gauges, MEMS accelerometers, LVDT/laser displacement, stable DAQ (prefer cabled), adhesives/consumables, safety PPE [6,7].

Software: MATLAB (Signal Processing, Control System, Optimization as required). Version control and data logging.

Facilities: City Campus structures lab; bench space; safe load equipment and fixtures.

C. Design Method

C1. Sensor Suite and Placement Rationale (RQ1)

Strain (primary): bottom chord mid-span (gravity tension), selected diagonals with greatest axial influence, and at least one top-chord compression member for buckling reserve [7,17].

Acceleration (global modes): mid-span deck/diaphragm and a quarter-point to characterise the first bending modes [6,12].

Displacement (serviceability): single mid-span LVDT/laser to constrain global stiffness and serve as a drift/thermal reference [7].

Placement is guided by Fisher-information/parameter sensitivity and mode observability which is mixing OSP theory with lab practicality [4,8,9].

C2. Data Acquisition and Quality Assurance

Sampling: 100–200 Hz for accel/strain (sufficient for low-frequency truss dynamics); anti-aliasing; synchronisation via common clock or timestamp alignment.

QA checks: range/variance monitors, stuck-value detection, hold-last-good logic for short dropouts.

Pre-processing: temperature compensation (strain), high-pass filtering for accelerations, moving-median outlier suppression [6,7].

C3. Physics Core and Estimation Algorithms (RQ2–RQ3)

Truss solver: linear-elastic 2-D FE (or analytical); local–global assembly; compute nodal displacements and axial forces each cycle [17].

State estimation: discrete Kalman filter for displacement/velocity states; measurement vector contains mid-span displacement and modal-filtered accelerations [11,12].

Parameter updating recursive least squares (or Gauss–Newton) on selected stiffness scalars; minimise residuals between predicted and measured strains/deflections [10].

Outputs: member axial-force histories, residual maps, uncertainty bands, deflection plots, and warnings for out-of-family behaviour.

C4. Calibration and Validation (RQ3)

Calibration: quasi-static step loads at safe levels; adjust stiffness scalars/boundary modifiers; target $\leq 10\%$ RMS error at service loads [10,11,17].

Validation: hold-out experiments; synthetic “damage” via virtual stiffness reductions in the model. Compute localisation (precision/recall), force-prediction error, and alarm thresholds.

Failure-point inference: use influence lines from the calibrated model to identify members at risk under moving loads; check yield/buckling capacity per code guidance [7,17].

C5. Performance Metrics and Acceptance Criteria

Static accuracy: deflection and strain RMS error (target: $\leq 10\%$ at service load).

Dynamic fidelity: first/second frequency error $\leq 5\%$; MAC with OMA mode shapes ≥ 0.9 where applicable [12].

Localisation: precision/recall ≥ 0.8 for synthetic stiffness losses $\geq 20\%$ in instrumented panels.

Latency: end-to-end cycle time < 150 ms for quasi-static tracking.

Robustness: graceful degradation with single-sensor dropout; bounded divergence with re-initialisation.

C6. Deliverables (Capstone A)

- Architecture and sensor placement justification.
- MATLAB twin pipeline (acquisition broker + solver + estimator stubs).
- Calibration/validation plan and metrics.
- Test protocols and SOPs.
- Part-B implementation roadmap.

D. Alternative Design

Sensor mix variants: (1) strain-centric (denser gauges, minimal dynamics) for strongest force inference; (2) dynamics-centric (more accelerometers, modal tracking) for global change detection.

Estimation schemes: (1) fixed-parameter state estimation for maximum stability; (2) dual estimation (parameters + states) for flexibility; (3) batch update offline with light online correction [10–12].

Model granularity: analytical truss (light, transparent) vs coarse FE with joint flexibility (higher fidelity, heavier).

Compute/deployment: low-latency all-local MATLAB vs edge + cloud storage/analysis (delayed).

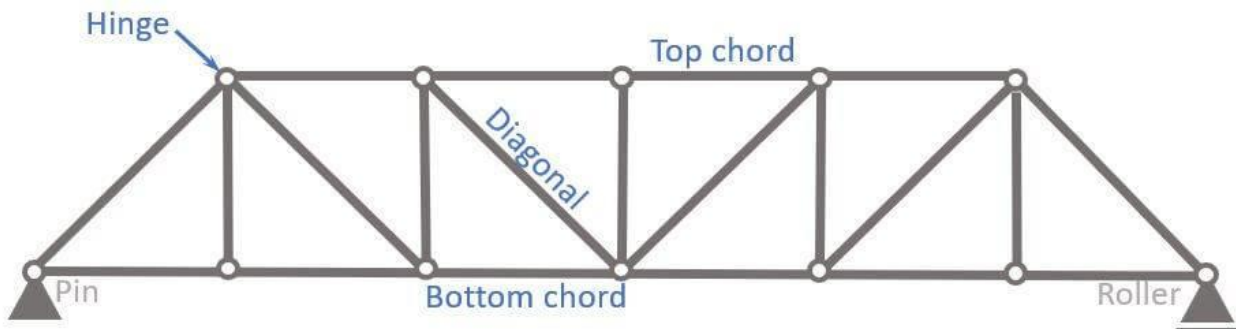


Figure 3. The Pratt truss is a commonly used type of truss structure shown with the member names.

Source: Structural Basics [18]

Risk Assessment / Team Governance:

At this stage, there is no planned laboratory attendance; accordingly, no risk assessment is provided. However, potential challenges include project delays, unclear task allocation, uneven workload distribution, and issues with file and version management.

Roles (five-person team):

- **Sensing Lead:** Choose sensors and their locations; prepare a simple installation guide.
- **Modelling Lead:** Create truss model and perform initial checks (supports, members).
- **Software Lead:** integrate MATLAB data pipeline, implement filters and estimators, and maintain code quality.
- **QA/Validation Lead:** Define and execute standard checks (e.g., error, latency) and grant final approval prior to major submissions.
- **PM/Docs Lead:** Keep schedule, keep decision log, manage versions, submit work.

Collaboration:

- **Meetings:** 30-min weekly stand-up + one short mid-week update in chat/doc.
- **Quality gate:** No merge/submit without peer review, functional code, and short README.
- **Changes:** Document scope changes; negotiate the impact (time/quality) with the supervisor beforehand.
- **Files:** One shared repo/folder; descriptive filenames; back up data and code.

If someone isn't collaborating:

- Mention it early in the weekly meeting.
- Develop a rescue plan (pair up, assign a smaller task, and set a new deadline).
- If more than twice, report to supervisor, delegate urgent tasks, and note contributions to the group table.

Ethical Considerations

- **Safety and Duty of Care:** All procedures follow lab SOPs and supervisor approvals; loads within conservative limits; exclusion zones enforced.
- **Academic Integrity:** No fabrication of data; assumptions and limitations disclosed; numbered references used [4,5].
- **Data Governance:** Controlled file naming/versioning; secure storage for raw/processed data; documented provenance.
- **Transparency and Accountability:** Weekly notes; decision logs; clear authorship and contribution records aligned with program guidelines.
- **Sustainability and Responsibility:** Minimise consumables; reuse sensors/adhesives where safe; power down equipment when idle.
- **Equity and Team Professionalism:** Respectful collaboration; task allocation matched to skills; proactive communication of risks and blockers.

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